position made it irritatingly difficult for pilots to climb into the cockpit, and they were alarmingly difficult to land, since the pilot had to make the vertical descent backward with his feet above his head. The unsatisfactory nature of these aircraft led to experiments with installing extra lift engines to otherwise conventional aircraft. Dassault equipped the Mirage III V with eight small engines to raise it vertically into the air, after which it flew conventionally. But the extra engines had a severe adverse effect on the aircraft's overall performance.

Jump Jet

The solution was found in the 1960s with the British Aerospace Harrier "Jump Jet," initially developed by Hawker Siddeley. It had the same jet engine for vertical takeoff and landing and for conventional flight, using swiveling nozzles to direct the thrust either down toward the ground or toward the rear of the aircraft. It could also take off conventionally from a short runway or flight deck. The main function originally envisaged for the Harrier was as a battlefield-support aircraft flying from concealed sites just behind the front line. In part, its thunder was stolen by the development of helicopter gunships, which compete for the same role and also do not need airstrips. The Harrier found its uses in some specific circumstances, and was astonishingly ingenious, but remained essentially marginal. Vertical takeoff was simply not compatible with the highest levels of performance in speed, range, or payload. Military planners were stuck with their airfields.

> "I'm not allowed to say how many planes joined the raid, but I counted them all out, and I counted them all back."
>
> **Brian Hanrahan**
> Reporting on a Harrier raid in the Falklands War

The Falklands War

The Harrier had its moment of glory in the spectacular but idiosyncratic air-sea encounters of the Falklands War, provoked by Argentina's occupation of the British Falkland Islands in 1982. The Royal Navy was sent to the stormy South Atlantic without either a full-size aircraft carrier or an airborne early-warning aircraft – by then an essential element of fleet air defense. Against the British Task Force, defended by a score of Sea Harriers and RAF Harrier GR.3s, the Argentinians deployed aircraft of 1960s vintage, mostly Mirage IIIs and A4 Skyhawks, plus a handful of Super Etendards.

Early in the conflict, an RAF Vulcan became the only member of Britain's one-time independent strategic nuclear bomber force to be used in anger, flying a 3,900-mile (6,250-km) roundtrip to bomb an airstrip on the Falklands. Delivered by the Super Etendards, the Exocet, an unexceptional French antiship missile, leapt to fame through sinking two British vessels. And the Sea Harriers and GR.3s, with their highly trained Navy and RAF pilots, proved more than a match for the aging Argentinian jets. Even more impressive than their combat performance was their ability to maintain constant combat air patrols in often dreadful weather conditions.

The main lesson of the Falklands War was, once again, the vital importance of electronic warfare and missile technology. It was the absence of an early-warning aircraft that made the Royal Navy's ships vulnerable to the Exocet missile. And